



GATE CS / IT 2027

Engineering Mathematics

Quick Revision Sheet

Complete GATE syllabus coverage — All topics and subtopics

GATEFlow Preparation Platform

Mathematical Logic**Propositional Logic**

- Connectives: NOT, AND, OR, IMPLIES, IFF
- Tautology, Contradiction, Contingency
- De Morgan Laws: $\text{NOT}(P \text{ AND } Q) = \text{NOT } P \text{ OR NOT } Q$
- Precedence: NOT > AND > OR > IMPLIES > IFF

First Order Logic

- Universal quantifier: for all x $P(x)$
- Existential quantifier: there exists x $P(x)$
- Rules of Inference: Modus Ponens, Modus Tollens, Resolution
- Predicate logic extends propositional with quantifiers

Set Theory & Algebra**Sets & Relations**

- Power Set of A with n elements has 2^n elements
- Cartesian Product: $|A \times B| = |A| \times |B|$
- Equivalence Relation: Reflexive + Symmetric + Transitive
- Partial Order: Reflexive + Antisymmetric + Transitive
- POSET, Hasse Diagrams, Lattices

Functions

- Injective (1-1): $f(a)=f(b)$ implies $a=b$
- Surjective (onto): every element of codomain is mapped
- Bijective = Injective + Surjective
- Inverse function exists iff function is bijective

Combinatorics**Counting**

- Permutations: $nPr = n!/(n-r)!$
- Combinations: $nCr = n!/r!(n-r)!$
- Inclusion-Exclusion: $|A \cup B| = |A| + |B| - |A \cap B|$
- Pigeonhole: $n+1$ items in n boxes $\Rightarrow$ 1 box has ≥ 2

Recurrences

- Master Theorem: $T(n) = aT(n/b) + f(n)$
- Case 1: $f(n) = O(n^{(\log_b(a)-\epsilon)}) \Rightarrow T(n) = \Theta(n^{\log_b(a)})$
- Case 2: $f(n) = \Theta(n^{\log_b(a)}) \Rightarrow T(n) = \Theta(n^{\log_b(a)} \log n)$
- Case 3: $f(n) = \Omega(n^{(\log_b(a)+\epsilon)}) \Rightarrow T(n) = \Theta(f(n))$

Probability

Probability Rules

- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
- $P(A|B) = P(A \cap B) / P(B)$
- Bayes Theorem: $P(A|B) = P(B|A) * P(A) / P(B)$
- Independent events: $P(A \cap B) = P(A) * P(B)$

Distributions

- Binomial: $P(X=k) = C(n,k) * p^k * (1-p)^{(n-k)}$
- Expected value $E[X] = \sum x * P(x)$
- Variance $Var(X) = E[X^2] - (E[X])^2$
- Normal distribution: mean= μ , std dev= σ

Linear Algebra

Matrices

- Rank = number of linearly independent rows/columns
- $\det(AB) = \det(A) * \det(B)$
- Inverse exists iff $\det(A) \neq 0$
- Rank-Nullity: $\text{rank}(A) + \text{nullity}(A) = n$ (columns)

Eigenvalues

- Characteristic equation: $\det(A - \lambda I) = 0$
- Sum of eigenvalues = $\text{Trace}(A)$
- Product of eigenvalues = $\det(A)$
- Cayley-Hamilton: matrix satisfies its own characteristic equation

Graph Theory

Key Theorems

- Handshaking Lemma: Sum of degrees = $2|E|$
- Tree: connected, acyclic, $|E| = |V| - 1$
- Euler Circuit: all vertices have even degree
- Hamilton Path: visits every vertex exactly once
- Planar Graph: $|E| \leq 3|V| - 6$ (Euler formula $V - E + F = 2$)
- Complete graph K_n has $n(n-1)/2$ edges